



Particle swarm optimization with problem-aware hyperparameter design for feature selection in high dimensions

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ABSTRACT

Feature selection is a data dimensionality reduction method commonly used to handle high-dimensional datasets. It improves classification accuracy by selecting relevant features, as well as removing irrelevant and redundant ones. However, due to the vast search space, effectively selecting features remains a challenge. Particle swarm optimization (PSO) has been successfully applied to feature selection due to its excellent global search capability and ease of implementation. Existing PSO methods typically use predefined or self-adaptive inertia weights and predefined acceleration coefficients for all datasets. The diversity across datasets means that uniform hyperparameter settings are often suboptimal. To address this, this paper introduces a novel PSO variant, termed PAPSO, which emphasizes a problem-aware hyperparameter design for improved adaptability. Specifically, PAPSO leverages each dataset's intrinsic properties via two mechanisms: an inertia weight adjustment driven by PSO's optimization process for tailored search behavior, and an initialization strategy that sets acceleration coefficients based on the dataset's specific characteristics. This data-driven approach is central to PAPSO's problem-aware capabilities, allowing the datasets's specific properties themselves to directly guide the hyperparameter design and optimization process more effectively than predefined settings. Extensive experiments on 15 high-dimensional datasets demonstrate that PAPSO achieves a superior balance between the number of selected features and classification accuracy.

1. Introduction

High-dimensional datasets are increasingly common in classification tasks. However, not all features in these datasets contribute meaningfully to classification performance. The inclusion of irrelevant or redundant features can undermine both accuracy and computational efficiency [1]. To tackle this issue, feature selection (FS) has emerged as a powerful technique to enhance model performance by identifying and retaining only the most informative features [2].

FS is a widely adopted dimensionality reduction strategy in data mining. Unlike feature extraction [3], which generates new features derived from the originals to reduce dimensionality, FS directly selects the most relevant features while eliminating those

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that are irrelevant or redundant. This approach not only boosts classification accuracy and computational efficiency but also maintains the interpretability of the original feature space [4].

Based on evaluation criteria, FS methods can be categorized into filter methods, embedded methods, and wrapper methods. Filter methods select a fixed number of top-ranked features based on statistical or information-theoretic measures. Since they operate independently of any specific learning algorithm, they are computationally efficient [5]. However, a major limitation of filter methods is their inability to capture feature dependencies, as features are evaluated individually rather than in combination [6]. Embedded methods integrate FS into the model training process, where FS is guided by the model's learning mechanism. Although embedded methods often outperform filter methods in terms of predictive performance, their selection criteria are typically model-specific, which may limit their applicability across different learning frameworks [7]. Wrapper methods, in contrast, evaluate feature subsets by incorporating a machine learning model into the selection process. These methods typically employ search strategies, such as evolutionary algorithms [8], to explore the feature space. By directly optimizing model performance, wrapper methods often achieve superior classification accuracy compared to filter and embedded methods. However, their effectiveness is highly dependent on the search strategy used. Thus, designing an efficient and effective search strategy is crucial to improving the performance of wrapper-based FS methods [9].

Evolutionary algorithms (EAs) [10], as a type of wrapper method, have demonstrated strong performance in feature selection (FS) tasks, particularly in high-dimensional settings. Among the various EAs, particle swarm optimization (PSO) has been widely applied due to its robust global search capability [11]. Recent research has yielded a variety of advanced algorithms to address the challenges of high-dimensional FS. For instance, some studies focus on innovations in related EAs domains. Yu et al. [12] propose a multi-population differential evolution algorithm for FS, which maintains three interacting populations to share information and incorporates an individual enhancement strategy to preserve population diversity. Shang et al. [13] propose a hybrid algorithm that combines the grey wolf optimizer and genetic algorithm; this method leverages chaotic maps and opposition-based learning for population initialization, and incorporates an elite learning strategy to enhance search efficiency. Concurrently, the PSO field has also seen significant progress. Jin et al. [14] introduce APPSO, which employs an adaptive constrained updating strategy to enhance exploration and a dynamic flip mechanism to improve exploitation. Yang et al. [15] propose LAPSO-DR, a leader-adaptive PSO with a dimensionality reduction strategy, leveraging a hybrid initialization approach and adaptive mechanisms. Another study, EBPSO [16], uses principal component analysis for dimensionality reduction and a novel position-update mechanism to avoid local optima traps. Furthermore, Gupta et al. [17] propose FPSO with several search mechanisms, including a weighted center-based approach, historical information-based acceleration coefficients, and a selection operation. Despite advances in applying PSO to FS, the performance of many methods remains sensitive to the manual setting of crucial parameters like inertia weight and acceleration coefficients [18]. These predefined parameters often lack adaptability to the unique characteristics of different datasets, potentially hindering optimization performance. To overcome this limitation, we propose PAPSOP, an effective PSO-based FS method emphasizing problem-aware hyperparameter design. Unlike traditional approaches, PAPSOP determines its key hyperparameters by leveraging feature importance information extracted directly from the dataset. This tailored approach enhances the algorithm's adaptability and effectiveness across diverse high-dimensional data. The primary contributions of this work are outlined as follows:

- (1) We propose a problem-aware inertia weight mechanism, where the inertia weight is dynamically determined by separately evaluating each feature's importance based on partial instances and importance based on all instances. Two optimizable parameters are introduced to represent these factors, and their combination adjusts the inertia weight during the search. Additionally, these parameters are incorporated into the PSO optimization process, enabling automatic refinement and improved search efficiency.
- (2) We propose an acceleration coefficient initialization mechanism for PAPSOP, wherein the coefficients are derived from the statistical properties of diverse feature importance measures. By assigning unique coefficients to each feature, this approach enhances the convergence behavior of the algorithm, improving its efficiency and adaptability across datasets.

The remaining of this paper are organized as follows: Section 2 provides the background of this paper, covering the concepts of feature selection, PSO, as well as ReliefF and mutual information (MI). In Section 3, we present a detailed description of the three key mechanisms of PAPSOP. Sections 4 and 5 describe the experimental setup and results, respectively, while Section 6 concludes the paper.

2. Background and related work

2.1. Feature selection

FS is a challenging task due to the vast search space it entails [19]. For a dataset comprising D features, there exist 2^D possible feature subsets that must be evaluated. The primary objective of FS is to enhance model performance while simultaneously reducing computational costs. Within this framework, FS aims to identify the smallest subset of features that maximizes classification accuracy, denoted by $M(S)$. This optimization problem can be formally expressed as follows:

$$\begin{aligned}
 &\max \quad M(S) \\
 &\text{s.t. } S = \{s_1, s_2, \dots, s_D\} \\
 &\quad s_i \in \{0, 1\} \quad i = 1, 2, \dots, D
 \end{aligned} \tag{1}$$

where S represents the selected feature subset, and s_i indicates the state of the i -th feature. Specifically, $s_i = 0$ implies that the i -th feature is excluded from the subset, while $s_i = 1$ indicates its inclusion.

2.2. Particle swarm optimization

PSO is a swarm intelligence algorithm inspired by social behaviors observed in nature, such as flocking birds and schooling fish [20]. In PSO, each potential solution to an optimization problem is represented by a particle within the swarm. Each particle maintains a position and velocity, which are iteratively updated to explore the search space. The swarm collectively tracks a global best position ($Gbest$), representing the best solution identified by any particle. Additionally, each particle records its personal best position ($Pbest$), corresponding to the best solution it has individually encountered. The movement of each particle is influenced by both its $Pbest$ and $Gbest$, facilitating convergence toward an optimal solution. The update equations are defined as follows:

$$\begin{aligned} v_i^{t+1} &= w \cdot v_i^t + c_1 \cdot r_1 \cdot (Pbest_i - x_i^t) + c_2 \cdot r_2 \cdot (Gbest - x_i^t) \\ x_i^{t+1} &= x_i^t + v_i^{t+1} \end{aligned} \quad (2)$$

where v_i^t represents the velocity of i -th particle at the t iteration, and x_i^t represents its position. The parameter w is the inertia weight, while c_1 and c_2 are acceleration coefficients that control the influence of $Pbest$ and $Gbest$, respectively. The terms r_1 and r_2 are randomly generated values in the range $[0, 1]$, introducing stochasticity into the update process to enhance exploration.

2.3. ReliefF

The ReliefF algorithm is an advanced feature evaluation technique that utilizes local neighborhood information to assess feature importance [21]. It operates by randomly selecting a sample l_i from the dataset and identifying its k nearest neighbors from the same class, referred to as same-class neighbors $H_{i,h}$ (where $h = 1, 2, \dots, k$), and k nearest neighbors from opposite classes, referred to as different-class neighbors $M_{i,p}$ (where $p = 1, 2, \dots, k$). Feature weights are updated by comparing the differences in feature values between the selected sample and these neighbors, with contributions averaged across the k neighbors to enhance robustness against noise and multi-class scenarios. The weight update rule is given by:

$$R(j) = R(j) - \frac{1}{m \cdot k} \sum_{i=1}^m \sum_{h=1}^k \text{diff}(l_{i,j}, H_{i,h,j}) + \frac{1}{m \cdot k} \sum_{i=1}^m \sum_{p=1}^k \text{diff}(l_{i,j}, M_{i,p,j}) \quad (3)$$

where $R(j)$ denotes the weight of feature j , reflecting its importance; m represents the number of randomly sampled instances; and k is the number of nearest neighbors considered. The term $l_{i,j}$ indicates the value of the j -th feature for the i -th sample, while $H_{i,h,j}$ and $M_{i,m,j}$ represent the values of the h -th same-class neighbor and m -th opposite-class neighbor, respectively, for sample l_i on feature j . The function $\text{diff}(\cdot)$ computes the normalized difference between two values. By relying on randomly sampled instances and their local neighbors, ReliefF emphasizes the discriminative power of features within partial, localized subsets of the dataset.

2.4. Mutual information

MI is an information-theoretic measure used to quantify the amount of information shared between two random variables [22]. In the context of feature selection, MI measures the dependency between a feature A and the target variable B . The definition is given by the following equation:

$$I(A; B) = \sum_{a \in A} \sum_{b \in B} p(a, b) \log \frac{p(a, b)}{p(a)p(b)} \quad (4)$$

where $I(A; B)$ is the MI between feature A and target variable B . $p(a, b)$ denotes the joint probability distribution between A and B , while $p(a)$ and $p(b)$ represent the marginal probability distributions of A and B , respectively. Since MI is calculated using all samples in the dataset, it captures the overall dependency between features and the target variable, whether this dependency is linear or nonlinear.

2.5. Existing PSO-based methods for FS

Due to its robust global search capability and ease of implementation, PSO has been widely applied to feature selection problems. Consequently, numerous PSO variants have been proposed for feature selection, demonstrating remarkable performance. For instance, Chen et al. [23] proposed a PSO variant called CUS-SPSO, which employs a ReliefF-based strategy to generate more promising solutions and utilizes a surrogate model to further enhance its performance. Li et al. [24] proposed ISBPSO, which uses mutual information for initialization, reduces the search space via a masking strategy, and incorporates genetic operations to avoid premature convergence. Another study [25] utilizes ReliefF to initialize two separate populations and employs them to search for solutions concurrently. While these studies have achieved considerable success, they reveal two common limitations. First, they often rely on predefined or generically adaptive hyperparameters, a setup that lacks adaptability to the unique characteristics of different datasets. Second, they typically leverage only a single source of prior knowledge to evaluate features. Given the complexity of feature relationships [26], relying on a single metric can introduce bias and provide an incomplete assessment of feature importance. To address these

Algorithm 1: The pseudo-code of PAPSO.

```

1 Calculate the features' ReliefF and MI and normalize them;
2 Generate a random population of size  $(N, D + 2)$ ;
3 Update the  $P_{best}$  and  $G_{best}$ ;
4 Initialize the acceleration coefficients;
5 while  $t < T$  do
6   for  $i = 1 : N$  do
7     Calculate the inertia weight based on (5);
8     for  $j = 3 : D + 2$  do
9       Update the  $x_{i,j}$  according to (7);
10    end
11    for  $j = 1 : 2$  do
12      Update the  $\alpha_j$  using (8);
13    end
14    Update the  $P_{best}$  and  $G_{best}$ ;
15  end
16   $t = t + 1$ ;
17 end

```

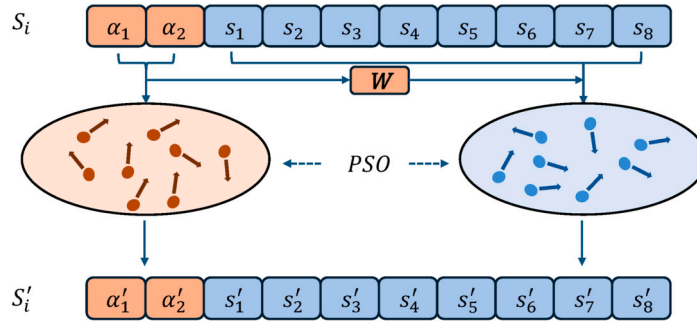


Fig. 1. Overview of inertia weight adjustment via PSO.

interconnected limitations, this paper proposes PAPSO, a novel PSO algorithm featuring a problem-aware hyperparameter design that utilizes both ReliefF and MI to dynamically guide the search process.

3. Methodology

In this section, we provide an explanation of PAPSO. As shown in Algorithm 1, after computing and normalizing the ReliefF and MI values for each feature, PAPSO augments the original PSO particle dimensions by adding two extra dimensions, thereby initializing a population of size $(N, D + 2)$ at random. During the iterative process, for each particle, PAPSO computes the inertia weight based on the first two dimensions of the particle, and subsequently updates these two dimensions after updating the remaining components. In the following sections, we will elaborate on the PAPSO algorithm in detail.

3.1. Inertia weight adjustment via PSO

In traditional PSO, the inertia weight w is typically a fixed value or dynamically adjusted based on predefined strategies, such as linearly decreasing with the number of iterations [27]. However, these approaches have their limitations, as it cannot fully adapt to the varying importance of features across different datasets and tasks. To overcome this limitation, we propose a problem-aware inertia weight that incorporates two different feature evaluation metrics: ReliefF and MI derived from datasets. ReliefF evaluates feature importance using randomly selected instances, focusing on local discriminative power, while MI utilizes all instances to compute the global dependency between features and the target variable. By combining these metrics, we aim to construct an inertia weight w that adjusts the search behavior of PSO to focus on features with balanced local and global significance. Specifically, we define the inertia weight as follows:

$$w(j) = \alpha_1 \cdot R(j) + \alpha_2 \cdot I(j) \quad (5)$$

where, α_1 and α_2 are optimization parameters, $R(j)$ represents the normalized ReliefF value of feature j , and $I(j)$ denotes the normalized MI between the target and feature j . This formulation adaptively balances the contributions of the ReliefF criterion and the MI criterion, thereby effectively guiding the search process within the PSO framework. By assigning an inertia weight to each feature, the algorithm can apply different inertial effects, thereby enhancing its search capability. A natural question arises: how can α_1 and α_2 be optimized? In this work, we address this by employing PSO itself to fine-tune these parameters, leveraging the algorithm's inherent optimization capabilities. Specifically, as shown in Fig. 1, the orange part represents the optimization of

Algorithm 2: Acceleration coefficient initialization.

Input: $R; I; c_{max}; c_{min}; D$
Output: c_1, c_2
1 Calculate the mean and standard deviation of ReliefF and MI, yielding $\mu_{reliefF}, \sigma_{reliefF}$ and μ_{mi}, σ_{mi} respectively.
2 **for** $j = 1$ to D **do**
3 Calculate the original $c_1(j)$ and $c_2(j)$ based on (6).
4 **end**
5 Use min-max normalization, c_1 and c_2 are mapped to the interval $[c_{min}, c_{max}]$.

the inertia weight parameters (α_1, α_2), while the blue part represents the optimization of the feature subset. We extend the search space of PSO by incorporating two additional dimensions, α_1 and α_2 , into the position and velocity vectors of each particle. These dimensions are used to dynamically adjust the inertia weight. Together with the traditional feature selection variables, they form the particle's solution space. During the iterative process of PSO, particles simultaneously update their positions in both the feature selection space and the parameter space, thus achieving joint optimization of both the feature subset and the parameters.

3.2. Acceleration coefficient initialization

Similar to the original inertia weight, the acceleration coefficients in PSO—typically set to fixed values—often require adaptation to different datasets [16]. In feature selection tasks, where the disparities in feature importance are pronounced, fixed values may not adequately reflect the relative significance among features. To better steer the PSO search within the feature space, we propose an initialization method for c_1 and c_2 that is based on feature importance scores: c_1 is based on ReliefF for its local sensitivity and c_2 is based on MI for its global perspective. The specific method is shown in Algorithm 2. The initialization begins by computing the mean ($\mu_{reliefF}, \sigma_{reliefF}$) and standard deviation (μ_{mi}, σ_{mi}) of the normalized ReliefF and MI values, respectively. For each feature j , the initial acceleration coefficients are calculated as follows:

$$\begin{aligned} c_1(j) &= R(j) - \mu_{reliefF} + \sigma_{reliefF} \\ c_2(j) &= I(j) - \mu_{mi} + \sigma_{mi} \end{aligned} \quad (6)$$

Here, subtracting the mean centers the feature scores around zero, mitigating bias due to overall scale differences, while adding the standard deviation amplifies the relative variability among features. This adjustment ensures that features with above-average importance receive greater influence in the optimization process, whereas those below average are assigned relatively lower weights. Subsequently, min-max normalization is applied to map the initial c_1 and c_2 values to the range $[c_{min}, c_{max}]$, as defined in the algorithm. c_{min}, c_{max} are equal to 0.5 and 3, respectively. This normalization preserves the relative differences among features while ensuring consistency across varying measurement scales.

3.3. Search operator

In this section, we will introduce the update method of PAPSO. In PSO, the velocity update relies on three key components: inertia velocity, personal best individual, and global best individual. However, these standard components may not be sufficient to fully explore the search space when dealing with complex high-dimensional optimization tasks. To overcome this limitation and enhance the search capability of PSO, we use the concept of random perturbation and incorporate it into the velocity update process to enhance the performance [28]. The specific PSO update equation is as follows:

$$\begin{aligned} V_{i,j}^{t+1} &= w(j) \cdot V_{i,j}^t + c_1(j) \cdot r_1 \cdot (Pbest_{i,j} - x_{i,j}^t) + c_2(j) \cdot r_2 \cdot (Gbest_j - x_{i,j}^t) - r_3 \cdot (T - x_{i,j}^t) \\ x_{i,j}^{t+1} &= \begin{cases} 1 & V_{i,j}^{t+1} > threshold \\ 0 & \text{otherwise} \end{cases} \end{aligned} \quad (7)$$

where T represents a binary random number (either 0 or 1), and r_3 is a random number in the range $[0, 1]$, similar to r_1 and r_2 . $threshold$ is the binary threshold which is set to 0.6 [29]. Moreover, since feature selection is a binary problem, PAPSO compares the computed velocity with a threshold to determine the final selection of features. This mechanism adjusts the velocity by subtracting a perturbation term based on the difference between the particle's current position and a randomly generated binary value. This binary value simulates potential feature selection decisions (i.e., selecting or not selecting a feature), thereby enhancing PSO's search capability. Additionally, the two optimization parameters α_1 and α_2 in Section 3.1 are also updated based on this mechanism. However, considering that they are specifically designed to optimize the inertia weight, we have removed the inertia weight term to prevent a feedback loop in which the control parameters would be influenced by the inertia weight they regulate, potentially destabilizing the algorithm's performance. The specific update formula is as follows:

$$\begin{aligned} \alpha_1 &= 2 \cdot r_1 \cdot (Pbest_{i,\alpha_1} - x_{i,\alpha_1}^t) + 2 \cdot r_2 \cdot (Gbest_{\alpha_1} - x_{i,\alpha_1}^t) - r_3 \cdot (T - x_{i,\alpha_1}^t) \\ \alpha_2 &= 2 \cdot r_1 \cdot (Pbest_{i,\alpha_2} - x_{i,\alpha_2}^t) + 2 \cdot r_2 \cdot (Gbest_{\alpha_2} - x_{i,\alpha_2}^t) - r_3 \cdot (T - x_{i,\alpha_2}^t) \end{aligned} \quad (8)$$

where $Pbest_{i,\alpha_1}$ is the value of the α_1 dimension in the personal best solution of the i -th individual. $Gbest_{\alpha_1}$ represents the corresponding value in the global best solution, and x_{i,α_1}^t denotes the current value in the α_1 dimension for the i -th solution at iteration t . The definitions for α_2 follow the same pattern, just swapping α_1 to α_2 .

3.4. Computational complexity analysis

According to Algorithm 1, before the main loop of PAPSO, computing the ReliefF and MI for each feature requires $O(D)$ time. Within the PAPSO main loop, updating each individual has a time complexity of $O(N \times D)$, while updating the first two dimensions requires $O(N \times 2)$. Thus, the overall time complexity of the main loop is $O(N \times (D + 2))$, which simplifies to $O(N \times D)$ after ignoring constant factors. Consequently, the time complexity of PAPSO is $O(N \times D)$.

3.5. Fitness function

The goal of FS is to minimize the classification error rate while selecting the fewest features. Based on this concept, in our experiments, we used the following fitness function that balances the classification error rate and the number of selected features:

$$fitness = \alpha \cdot error + \beta \cdot \frac{D_s}{D} \quad (9)$$

where D_s represents the number of features in the selected feature subset, and $error$ denotes the classification error rate corresponding to the feature subset. α and β are parameters that balance these two objectives. In our experiments, we follow the settings in [23], where $\alpha = 0.99$ and $\beta = 0.01$.

3.6. Motivation for integrating ReliefF and MI

In feature selection, various methods exist to identify the most relevant features and reduce redundancy [30]. However, feature importance is not a single dimension; it can manifest both in the general association of a feature with the target variable across the entire dataset (global importance) and in the feature's discriminative ability within specific data regions or through interactions with other features (local importance) [31]. To assess feature value more comprehensively, we choose to combine two representative and complementary metrics: ReliefF and MI. As stated in the literature [32], ReliefF's weight is an average based on local estimates in smaller parts of the instance subspace, essentially representing local importance. Its evaluation relies on distinguishing between nearest neighbor samples within a local neighborhood, allowing it to utilize local information to characterize data relevance [33]. MI, on the other hand, offers a global perspective based on information theory to evaluate feature importance [34]. Unlike ReliefF, its calculation depends on the probability distribution of variables over the entire dataset, reflecting the average association strength or overall correlation between the feature and the target variable [35]. Therefore, we opt to combine the local sensitivity of ReliefF with the global view of MI, aiming for a more comprehensive evaluation of feature importance.

Having established this goal, the question becomes how best to fuse these two distinct metrics. In this work, we propose using a weighted linear combination, as defined in Equation (5), for two primary reasons. First, a linear combination represents the most direct and interpretable method for integrating two scores, maintaining a clear and understandable relationship between the individual metrics and the final combined importance value. Second, and more crucially, we empower the algorithm to autonomously learn the optimal balance for any given dataset. This mechanism effectively mitigates the limitations of a predefined weighting scheme, allowing PAPSO to decide whether local or global characteristics are more critical for solving a specific problem.

Indeed, this fundamental principle of balancing local and global perspectives resonates deeply with the core philosophy of PSO itself. The cognitive parameter c_1 guides particles towards local search, while the social parameter c_2 guides them towards global search, echoing the characteristics of ReliefF (local focus) and MI (global focus), respectively. Furthermore, the inertia weight in PSO is crucial for balancing the algorithm's global exploration and local exploitation capabilities. We use the combined feature importance information derived from ReliefF and MI to dynamically adjust the inertia weight. This approach makes the PSO search process more intelligent and data-driven, improving optimization performance. The experimental results in Section 5.5 also further validate the effectiveness of this combination.

3.7. General remarks of proposed mechanisms

PAPSO advances PSO for high-dimensional feature selection through a problem-aware hyperparameter design strategy, effectively integrating dataset-specific insights from ReliefF and MI. One of its key techniques is inertia weight adjustment via PSO, where the crucial balance between ReliefF and MI contributions in the inertia weight calculation is dynamically tuned by the PSO algorithm itself. Complementing this, PAPSO employs dataset-specific initialization for acceleration coefficients, assigning unique initial $c_1(j)$ (based on ReliefF) and $c_2(j)$ (based on MI) values reflecting each feature's significance within the specific dataset. To further ensure robust search, a random perturbation mechanism is incorporated into the velocity update process, enhancing exploration and mitigating premature convergence. These synergistic components grant PAPSO superior adaptability across diverse datasets and robust performance in identifying optimal feature subsets.

Table 1
The details of Datasets.

No.	Name	Instances	Features	Classes
1	warpAR10P	130	2400	10
2	warpPIE10P	210	2420	10
3	PCMAC	1943	3289	2
4	Leukemia1	72	5327	3
5	madelonrand	60	5726	9
6	TOX_171	171	5748	4
7	Prostate_GE	102	5966	2
8	DLBCL	77	7070	2
9	ALL-AML-3	72	7129	3
10	ALLAML	72	7129	2
11	CNS	60	7130	2
12	nci9	60	9712	9
13	arcene	200	10000	2
14	11tumors2	174	12533	11
15	CML	28	12625	2

4. Experiments setup

4.1. Datasets

In our experiments, we employ 15 high-dimensional datasets that are widely used in feature selection research to evaluate the performance of the proposed algorithm. These datasets have been utilized in various previous studies [36,37], ensuring a fair and comprehensive comparison. The details of these datasets are provided in Table 1. The number of features in these datasets ranges from a minimum of 2,400 to a maximum of 12,625, while the number of classes varies from binary classification to multi-class problems with up to 11 classes. These datasets cover a diverse range of application scenarios, making them well-suited for assessing the effectiveness and robustness of the proposed approach.

4.2. Comparison algorithms

To evaluate the performance of our proposed algorithm, we select six evolutionary algorithms commonly used of FS tasks, including FPSO [17], SSDR [38], EBPSO [16], MGWO [36], SFE-PSO [37] and CUS-SPSO [23]. These algorithms were chosen for their recent advancements and proven efficacy in addressing high-dimensional FS challenges. Below, we outline their key features.

- (1) FPSO: FPSO enhances binary PSO for FS by introducing a weighted center-based guiding vector, adaptive acceleration coefficients based on historical success, a greedy selection step, and an arctan-based transfer function.
- (2) SSDR: This algorithm tackles high-dimensional datasets by partitioning the search space into seven subspaces. Each subspace is explored using the BRAO method, and the resulting feature subsets are subsequently combined for further optimization.
- (3) EBPSO: EBPSO mitigates the premature convergence often observed in PSO by introducing a novel position update mechanism, replacing the conventional velocity-based approach with a new position variable strategy.
- (4) MGWO: Tailored for high-dimensional FS, MGWO integrates the ReliefF metric with Copula entropy for population initialization. It further enhances performance through competitive guidance and differential evolution techniques.
- (5) SFE-PSO: This method combines the SFE algorithm, which employs non-selection and selection operators for global and local searches to reduce the search space, with PSO to optimize the selection of feature subsets in high-dimensional data.
- (6) CUS-SPSO: CUS-SPSO excels in high-dimensional FS by utilizing a ReliefF-based update strategy to generate promising solutions, augmented by a surrogate model to further improve its performance.

4.3. Description of other metrics

To more comprehensively assess the performance of the proposed algorithm, in addition to the fitness metric, five additional key performance metrics are introduced: classification accuracy, the number of selected features, precision, recall, and F1-score.

4.3.1. Classification accuracy

Classification accuracy quantifies the overall predictive capability of the classifier on the dataset. It serves as an intuitive and critical measure for evaluating the effectiveness of the feature selection process. It is defined as follows:

$$\text{Classification accuracy} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(y_i = \hat{y}_i) \quad (10)$$

where n denotes the total number of instances, y_i represents the true label of the i th instance, $\hat{y}_i$ denotes the predicted label for the i th instance, and $\mathbb{I} \cdot$ is the indicator function.

Table 2
Parameter settings.

Comparison algorithms	Parameter settings
FPSO	$w_{max} = 0.9, w_{min} = 0.1$
SSDR	$K = 5$
EBPSO	$c_1 = 0.5, c_2 = 1, w = 0.5, lb = 0, ub = 0.9$
MGWO	$F = 0.5$
SFE-PSO	$UR = 0.3, UR_{max} = 0.3, UR_{min} = 0.001, c_1 = 1.5, c_2 = 2, w = 0.5$
CUS-SPSO	$c_1 = c_2 = 1.49445, n_c = 2, A = 0.15, B = 0.05$

4.3.2. Number of selected features

A key goal of feature selection is to minimize dimensionality while preserving robust classification performance. A reduced number of selected features improves computational efficiency and enhances the model's generalization capability. If S represents the set of selected features, this metric is expressed as:

$$\text{Number of selected features} = |S| \quad (11)$$

4.3.3. Precision

Precision evaluates the quality of the positive predictions made by the classifier. It is defined as the ratio of true positive predictions to the total number of instances predicted as positive:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12)$$

where TP denotes the number of true positives and FP denotes the number of false positives. A high precision indicates that a majority of the positive predictions are correct, which is particularly important in scenarios where the positive class is underrepresented.

4.3.4. Recall

Recall measures the classifier's ability to identify all relevant positive instances. It is calculated as the ratio of true positive predictions to the total number of actual positive instances:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (13)$$

where FN denotes the number of false negatives. A high recall value implies that the classifier is effective in capturing most of the positive instances, which is critical when the cost of missing a positive instance is high.

4.3.5. F1-score

The F1-score, defined as the harmonic mean of precision and recall, provides a single measure that balances both the precision and recall of the classifier:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (14)$$

The harmonic mean penalizes extreme values, ensuring that both precision and recall are adequately balanced, which is particularly beneficial in cases with imbalanced class distributions.

4.4. Parameter settings

For the experimental setup, parameter settings for all comparison algorithms are listed in Table 2. Common parameters across algorithms include a population size of 50 and a maximum of 100 iterations. A k -nearest neighbors classifier ($k = 5$) serves as the evaluation model to assess the performance of selected feature subsets. For performance evaluation, the dataset is initially partitioned into a training set (70%) and a test set (30%). The entire feature selection process—including the 5-fold cross-validation for evaluating subsets and the calculation of any guiding metrics (such as ReliefF and MI)—is performed exclusively on the training set. The selected subset is then evaluated on the held-out test set. Each algorithm is independently executed 30 times on each dataset, and the average values of all evaluation metrics over these 30 runs are reported as the final results.

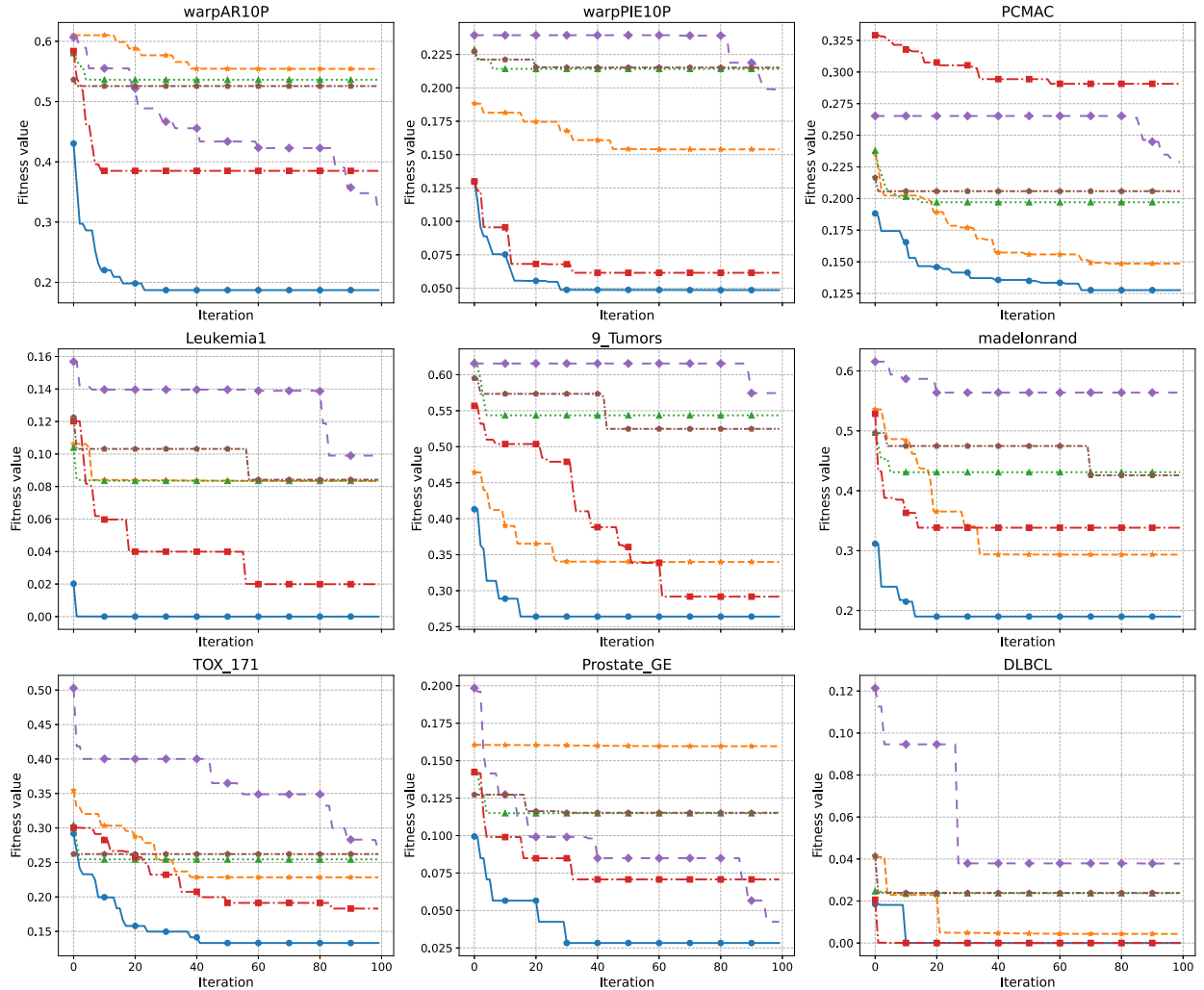
5. Experimental results

5.1. Convergence analysis

In this section, we compare the performance of PAPS0 with its peers in terms of fitness values. As detailed in Table 3, PAPS0 consistently converges to superior fitness values across all datasets, exhibiting a significantly lower average fitness value compared to the other comparative algorithms. To rigorously assess the statistical significance of these observations, we employ the Wilcoxon signed-rank test and Wilcoxon rank-sum test [39]. The Wilcoxon signed-rank test, conducted at a significance level of 0.05, aims to

Table 3Performance comparison across different algorithms with fitness values (mean $\pm$ std).

No.	CUS-SPSO (Chen et al. 2022)	SFE-PSO (Ahadzadeh et al. 2023)	MGWO (Pan et al. 2023)	EBPSO (Tijani et al. 2024)	SSDR (Chaudhuri 2024)	FPSO (Gupta et al. 2024)	PAPSO
1	0.538 $\pm$ 0.009(+)	0.351 $\pm$ 0.037 (+)	0.337 $\pm$ 0.052(+)	0.534 $\pm$ 0.017(+)	0.611 $\pm$ 0.015(+)	0.568 $\pm$ 0.013(+)	0.186 $\pm$ 0.024
2	0.212 $\pm$ 0.005(+)	0.193 $\pm$ 0.011(+)	0.047 $\pm$ 0.012(+)	0.214 $\pm$ 0.009(+)	0.180 $\pm$ 0.007(+)	0.163 $\pm$ 0.006(+)	0.037 $\pm$ 0.011
3	0.211 $\pm$ 0.007(+)	0.235 $\pm$ 0.031(+)	0.229 $\pm$ 0.041(+)	0.198 $\pm$ 0.018(+)	0.197 $\pm$ 0.037(+)	0.144 $\pm$ 0.010(+)	0.131 $\pm$ 0.011
4	0.072 $\pm$ 0.011(+)	0.043 $\pm$ 0.022(+)	0.057 $\pm$ 0.030(+)	0.080 $\pm$ 0.018(+)	0.068 $\pm$ 0.018(+)	0.053 $\pm$ 0.010(+)	0.003 $\pm$ 0.007
5	0.552 $\pm$ 0.018(+)	0.455 $\pm$ 0.050(+)	0.367 $\pm$ 0.051(+)	0.551 $\pm$ 0.032(+)	0.435 $\pm$ 0.028(+)	0.322 $\pm$ 0.027(+)	0.267 $\pm$ 0.036
6	0.445 $\pm$ 0.018(+)	0.399 $\pm$ 0.047(+)	0.346 $\pm$ 0.044(+)	0.436 $\pm$ 0.028(+)	0.387 $\pm$ 0.027(+)	0.306 $\pm$ 0.039(+)	0.187 $\pm$ 0.035
7	0.274 $\pm$ 0.009(+)	0.245 $\pm$ 0.043(+)	0.147 $\pm$ 0.021(+)	0.268 $\pm$ 0.024(+)	0.264 $\pm$ 0.016(+)	0.212 $\pm$ 0.014(+)	0.132 $\pm$ 0.024
8	0.111 $\pm$ 0.012(+)	0.040 $\pm$ 0.025(=)	0.062 $\pm$ 0.016(+)	0.115 $\pm$ 0.013(+)	0.148 $\pm$ 0.007(+)	0.152 $\pm$ 0.007(+)	0.035 $\pm$ 0.008
9	0.024 $\pm$ 0.000(+)	0.017 $\pm$ 0.017(+)	0.007 $\pm$ 0.011(+)	0.026 $\pm$ 0.006(+)	0.025 $\pm$ 0.009(+)	0.021 $\pm$ 0.009(+)	0.001 $\pm$ 0.003
10	0.038 $\pm$ 0.009(+)	0.027 $\pm$ 0.024(+)	0.067 $\pm$ 0.032(+)	0.050 $\pm$ 0.011(+)	0.084 $\pm$ 0.014(+)	0.083 $\pm$ 0.016(+)	0.016 $\pm$ 0.008
11	0.127 $\pm$ 0.012(+)	0.055 $\pm$ 0.050(+)	0.020 $\pm$ 0.013(+)	0.135 $\pm$ 0.018(+)	0.145 $\pm$ 0.004(+)	0.143 $\pm$ 0.009(+)	0.000 $\pm$ 0.000
12	0.136 $\pm$ 0.013(+)	0.118 $\pm$ 0.034(+)	0.142 $\pm$ 0.035(+)	0.146 $\pm$ 0.020(+)	0.186 $\pm$ 0.021(+)	0.116 $\pm$ 0.031(+)	0.059 $\pm$ 0.025
13	0.519 $\pm$ 0.009(+)	0.479 $\pm$ 0.025(+)	0.283 $\pm$ 0.058(+)	0.520 $\pm$ 0.022(+)	0.608 $\pm$ 0.017(+)	0.583 $\pm$ 0.017(+)	0.125 $\pm$ 0.039
14	0.168 $\pm$ 0.003(+)	0.148 $\pm$ 0.015(+)	0.081 $\pm$ 0.012(+)	0.167 $\pm$ 0.007(+)	0.148 $\pm$ 0.009(+)	0.137 $\pm$ 0.009(+)	0.074 $\pm$ 0.011
15	0.395 $\pm$ 0.025(+)	0.210 $\pm$ 0.096(+)	0.167 $\pm$ 0.064(+)	0.418 $\pm$ 0.030(+)	0.482 $\pm$ 0.031(+)	0.436 $\pm$ 0.039(+)	0.012 $\pm$ 0.021
Mean	0.255	0.201	0.157	0.257	0.265	0.229	0.084
p-value	3.052E-05	3.052E-05	3.052E-05	3.052E-05	3.052E-05	3.052E-05	—
w/t/l	15/0/0	14/1/0	15/0/0	15/0/0	15/0/0	15/0/0	—/—/—

**Fig. 2.** Convergence plots for all datasets.

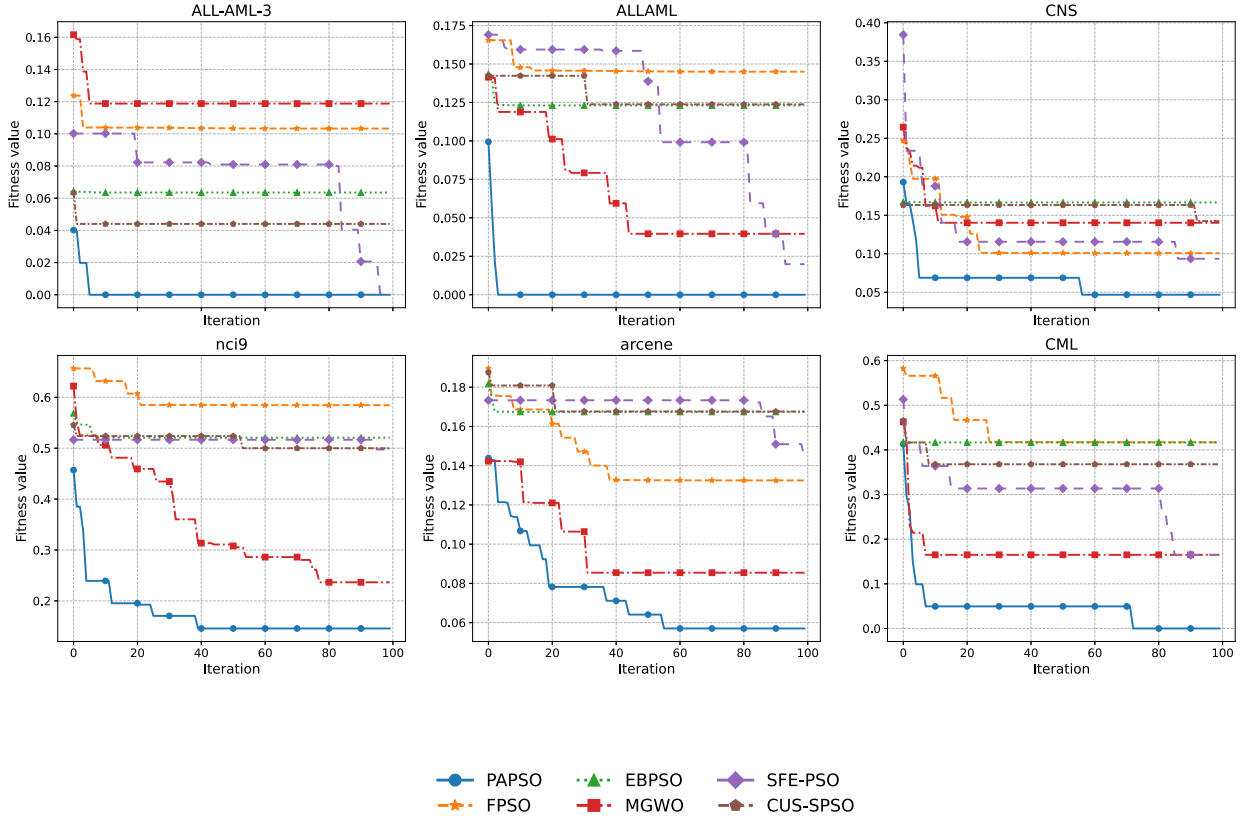


Fig. 2. (continued)

determine the overall statistical superiority of PAPSO across all datasets against its competitors. Complementarily, the Wilcoxon rank-sum test utilizes the symbols ‘+’, ‘=’ and ‘−’ to indicate whether PAPSO performs significantly better than, statistically equivalent to, or significantly worse than each competing algorithm on individual datasets. The results of both statistical tests strongly corroborate the exceptional convergence capability of PAPSO. Furthermore, the convergence behavior of all algorithms except SSDR is illustrated in Fig. 2, where the horizontal axis represents the number of iterations and the vertical axis displays the fitness values. Due to SSDR’s maintenance of multiple parallel subpopulations, depicting its convergence trajectory in a single figure presents a considerable challenge. As clearly shown in the convergence curves, PAPSO consistently reaches the minimum fitness value across all datasets, demonstrating a superior convergence speed by requiring notably fewer iterations to achieve the optimum compared to its peers. PAPSO’s rapid convergence is likely attributed to the data-driven initialization, which provides a more informed starting point for the search, and the inertia weight adjustment, which effectively balances exploration and exploitation throughout the optimization process. These findings unequivocally highlight PAPSO’s robust convergence capability and effectively validate the efficacy of the proposed approach.

5.2. Classification accuracy and feature size analysis

Given that classification accuracy and feature size are the primary objectives in feature selection, this section evaluates PAPSO’s performance by comparing its results on these two metrics against other algorithms. Detailed results are presented in Table 4. Notably, on datasets 5 and 11, PAPSO’s classification accuracy shows significant improvements compared to its competitors. Analyzing the average performance across all datasets, PAPSO demonstrates superior accuracy. Wilcoxon signed-rank tests further confirm that PAPSO is statistically superior in terms of accuracy compared to the other algorithms over the 15 datasets. Moreover, Wilcoxon rank-sum tests reveal that PAPSO outperforms EBPSO, SSDR, and FPSO in accuracy on at least 10 datasets. Specifically concerning MGWO, PAPSO achieves significantly superior accuracy on all 15 datasets. In terms of feature size, despite PAPSO selecting a larger number of features than MGWO, its classification accuracy significantly surpasses MGWO’s across all datasets. Furthermore, when compared to SSDR and FPSO – algorithms that exhibit comparable accuracy to PAPSO on some datasets – PAPSO selects considerably fewer features. For instance, on dataset 4, PAPSO utilizes approximately 1/200th the number of features selected by SSDR and FPSO. This stands in contrast to methods like SSDR and FPSO, which do not incorporate such prior knowledge about feature importance. Instead, PAPSO fully leverages the feature evaluations provided by ReliefF and MI. By using this information to adjust its inertia weight and acceleration coefficients, PAPSO better aligns the search process with the structure of each dataset. This targeted configuration enables

Table 4
Performance comparison across different algorithms with classification accuracy.

No.	EBPSO		SSDR		FPSO		PAPSO	
	Acc	Size	Acc	Size	Acc	Size	Acc	Size
1	0.429 ± 0.034(+)	1066.200	0.518 ± 0.030(+)	833.633	0.497 ± 0.032(+)	1017.467	0.825 ± 0.058	55.000
2	0.868 ± 0.021(+)	1086.867	0.865 ± 0.019(+)	854.067	0.864 ± 0.018(+)	994.900	0.893 ± 0.030	72.400
3	0.792 ± 0.016(+)	1625.933	0.801 ± 0.014(+)	1311.267	0.813 ± 0.016(+)	1638.467	0.825 ± 0.024	97.133
4	0.761 ± 0.062(=)	2330.267	0.830 ± 0.051(+)	2110.533	0.821 ± 0.049(+)	2277.633	0.900 ± 0.051	9.867
5	0.372 ± 0.101(+)	2624.867	0.394 ± 0.105(+)	2334.667	0.311 ± 0.065(+)	2558.300	0.454 ± 0.110	298.100
6	0.289 ± 0.079(+)	2645.033	0.298 ± 0.063(=)	2338.133	0.322 ± 0.074(=)	2622.767	0.326 ± 0.074	130.633
7	0.641 ± 0.059(+)	2765.000	0.725 ± 0.048(=)	2117.400	0.729 ± 0.054(=)	2722.367	0.747 ± 0.052	301.100
8	0.698 ± 0.020(+)	2604.700	0.937 ± 0.018(=)	1964.967	0.932 ± 0.016(=)	2482.267	0.941 ± 0.047	8.267
9	0.894 ± 0.026(=)	2963.333	0.887 ± 0.041(+)	2593.533	0.894 ± 0.037(=)	2980.900	0.907 ± 0.042	8.333
10	0.868 ± 0.038(=)	2964.267	0.802 ± 0.053(+)	2400.133	0.812 ± 0.043(+)	3042.300	0.874 ± 0.050	3.600
11	0.824 ± 0.016(+)	2921.133	0.824 ± 0.023(+)	2144.767	0.818 ± 0.000(+)	3006.300	0.939 ± 0.051	4.833
12	0.400 ± 0.049(+)	3155.600	0.639 ± 0.077(=)	2632.600	0.652 ± 0.062(=)	3227.900	0.657 ± 0.077	82.600
13	0.493 ± 0.063(+)	4323.267	0.439 ± 0.027(+)	3460.167	0.430 ± 0.025(+)	4332.033	0.528 ± 0.085	61.567
14	0.827 ± 0.018(+)	4599.967	0.854 ± 0.016(+)	3552.567	0.854 ± 0.013(+)	4637.167	0.878 ± 0.031	175.000
15	0.619 ± 0.063(+)	5197.733	0.504 ± 0.119(+)	4553.233	0.474 ± 0.092(+)	5686.300	0.685 ± 0.106	11.200
Mean	0.652	2857.878	0.688	2346.778	0.682	2881.871	0.759	87.976
p-value	3.052E-05	—	3.052E-05	—	3.052E-05	—	—	—
w/t/l	13/2/0	—/—/—	11/4/0	—/—/—	10/5/0	—/—/—	—/—/—	—/—/—

No.	CUS-SPSO		SFE-PSO		MGWO	
	Acc	Size	Acc	Size	Acc	Size
1	0.443 ± 0.028(+)	1063.033	0.539 ± 0.087(+)	60.400	0.553 ± 0.100(+)	9.800
2	0.872 ± 0.020(=)	1091.667	0.867 ± 0.019(+)	591.867	0.881 ± 0.023(+)	28.233
3	0.798 ± 0.018(+)	1588.800	0.744 ± 0.033(+)	62.867	0.699 ± 0.071(+)	82.867
4	0.780 ± 0.074(+)	2377.133	0.748 ± 0.077(+)	477.767	0.811 ± 0.109(+)	7.367
5	0.359 ± 0.106(+)	2688.333	0.300 ± 0.107(+)	944.200	0.337 ± 0.127(+)	55.567
6	0.302 ± 0.074(=)	2569.500	0.304 ± 0.106(+)	237.800	0.191 ± 0.068(+)	23.367
7	0.669 ± 0.042(+)	2599.867	0.634 ± 0.070(+)	148.567	0.686 ± 0.053(+)	122.233
8	0.689 ± 0.022(+)	2691.600	0.690 ± 0.078(+)	35.600	0.896 ± 0.074(+)	5.933
9	0.892 ± 0.023(+)	2837.933	0.889 ± 0.056(=)	176.733	0.874 ± 0.055(+)	14.600
10	0.859 ± 0.038(+)	3097.433	0.808 ± 0.068(+)	227.800	0.745 ± 0.113(+)	8.367
11	0.821 ± 0.012(+)	3130.600	0.824 ± 0.059(+)	608.933	0.812 ± 0.098(+)	6.567
12	0.380 ± 0.055(+)	3326.667	0.407 ± 0.084(+)	18.000	0.598 ± 0.096(+)	10.667
13	0.478 ± 0.069(+)	4602.400	0.496 ± 0.067(=)	2397.733	0.319 ± 0.120(+)	25.900
14	0.825 ± 0.020(+)	4616.500	0.822 ± 0.041(+)	2189.067	0.771 ± 0.053(+)	116.133
15	0.637 ± 0.077(+)	5347.467	0.604 ± 0.156(=)	416.433	0.578 ± 0.150(+)	2.967
Mean	0.654	2908.596	0.645	610.078	0.650	34.704
p-value	3.052E-05	—	3.052E-05	—	3.052E-05	—
w/t/l	13/2/0	—/—/—	12/3/0	—/—/—	15/0/0	—/—/—

Table 5
Performance comparison across different algorithms with precision.

No.	CUS-SPSO	SFE-PSO	MGWO	EBPSO	SSDR	FPSO	PAPSO
1	0.443 ± 0.028	0.539 ± 0.087	0.630 ± 0.087	0.429 ± 0.034	0.556 ± 0.038	0.528 ± 0.047	0.851 ± 0.058
2	0.872 ± 0.020	0.867 ± 0.019	0.895 ± 0.022	0.868 ± 0.021	0.870 ± 0.019	0.870 ± 0.020	0.901 ± 0.026
3	0.798 ± 0.018	0.744 ± 0.033	0.738 ± 0.060	0.792 ± 0.016	0.805 ± 0.014	0.814 ± 0.015	0.833 ± 0.026
4	0.780 ± 0.074	0.748 ± 0.077	0.824 ± 0.111	0.761 ± 0.062	0.865 ± 0.045	0.861 ± 0.044	0.913 ± 0.045
5	0.359 ± 0.10	0.300 ± 0.107	0.391 ± 0.132	0.372 ± 0.101	0.430 ± 0.144	0.280 ± 0.117	0.483 ± 0.141
6	0.302 ± 0.074	0.304 ± 0.106	0.211 ± 0.103	0.289 ± 0.079	0.364 ± 0.100	0.379 ± 0.104	0.362 ± 0.130
7	0.669 ± 0.042	0.634 ± 0.070	0.710 ± 0.052	0.641 ± 0.059	0.750 ± 0.041	0.750 ± 0.053	0.763 ± 0.051
8	0.689 ± 0.022	0.690 ± 0.078	0.904 ± 0.070	0.698 ± 0.020	0.943 ± 0.016	0.939 ± 0.015	0.946 ± 0.043
9	0.892 ± 0.038	0.889 ± 0.056	0.893 ± 0.049	0.894 ± 0.026	0.928 ± 0.018	0.931 ± 0.019	0.919 ± 0.046
10	0.859 ± 0.038	0.808 ± 0.068	0.764 ± 0.144	0.868 ± 0.038	0.812 ± 0.108	0.849 ± 0.057	0.896 ± 0.044
11	0.821 ± 0.012	0.824 ± 0.059	0.825 ± 0.094	0.824 ± 0.016	0.857 ± 0.027	0.856 ± 0.000	0.945 ± 0.047
12	0.380 ± 0.055	0.407 ± 0.084	0.583 ± 0.111	0.400 ± 0.049	0.623 ± 0.094	0.645 ± 0.069	0.657 ± 0.097
13	0.478 ± 0.069	0.496 ± 0.067	0.356 ± 0.145	0.493 ± 0.063	0.391 ± 0.016	0.390 ± 0.006	0.598 ± 0.122
14	0.825 ± 0.020	0.822 ± 0.041	0.779 ± 0.049	0.827 ± 0.018	0.855 ± 0.015	0.855 ± 0.12	0.880 ± 0.031
15	0.637 ± 0.077	0.604 ± 0.156	0.629 ± 0.122	0.619 ± 0.063	0.688 ± 0.132	0.698 ± 0.148	0.741 ± 0.131
Mean	0.654	0.645	0.676	0.652	0.716	0.710	0.779
p-value	3.052E-05	3.052E-05	3.052E-05	3.052E-05	2.136E-04	5.798E-04	—

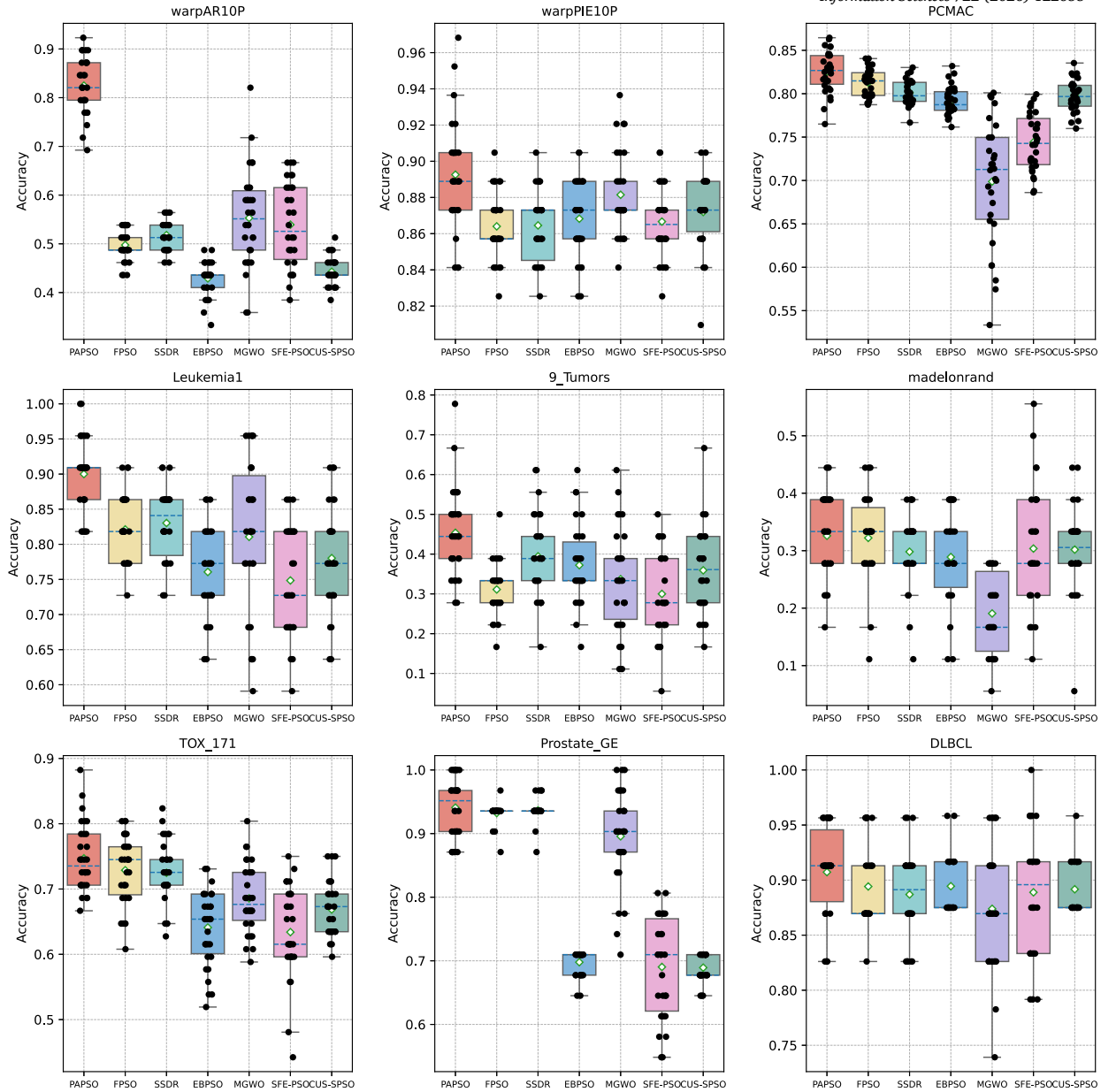


Fig. 3. Box-whisker-plots of all compared algorithm across all datasets.

a more balanced search process, improving the algorithm's ability to explore complex high-dimensional spaces and reducing the risk of premature convergence. As a result, PAPSO consistently identifies compact yet highly discriminative feature subsets, achieving a strong trade-off between feature quantity and predictive performance across diverse datasets.

In addition, we plot box-plots for all datasets, as shown in Fig. 3. It is evident that on 10 datasets, the maximum values achieved by PAPSO are optimal, and on 11 datasets, the median performance of PAPSO surpasses that of the other algorithms. These findings, combined with the statistical significance demonstrated by the Wilcoxon tests, further validate the robustness and effectiveness of PAPSO in FS tasks.

5.3. Other metrics analysis

To account for potential class imbalance in the datasets, in addition to classification accuracy, we also employ precision, recall, and F1-score as evaluation metrics, as they provide a more comprehensive assessment of performance. Detailed comparisons are summarized in Tables 5 to 7. Compared to other algorithms, PAPSO consistently achieves leading results, securing the highest average precision, recall, and F1-score across all datasets. Furthermore, Wilcoxon statistical tests confirm that these performance

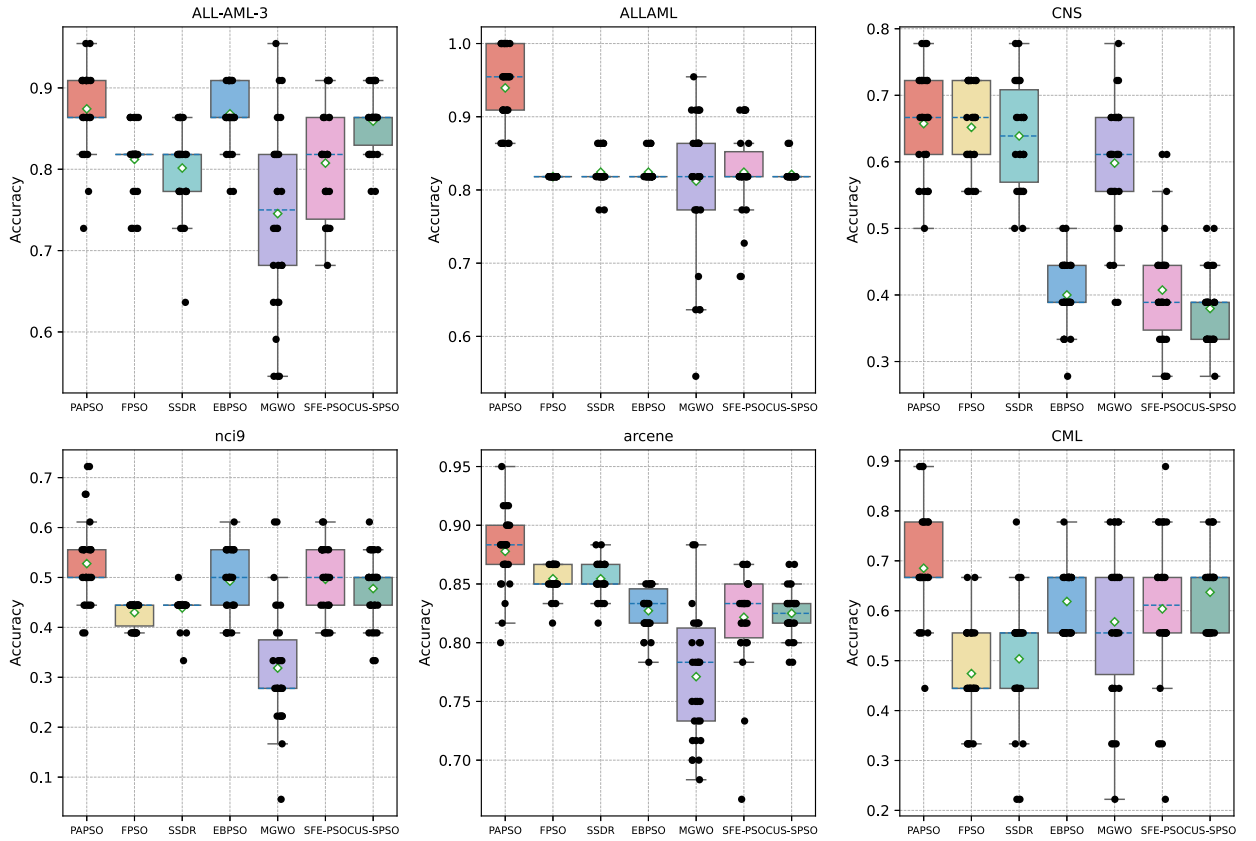


Fig. 3. (continued)

Table 6
Performance comparison across different algorithms with recall.

No.	CUS-SPSO	SFE-PSO	MGWO	EBPSO	SSDR	FPSO	PAPSO
1	0.439 ± 0.027	0.518 ± 0.090	0.553 ± 0.100	0.427 ± 0.038	0.518 ± 0.030	0.497 ± 0.032	0.825 ± 0.058
2	0.874 ± 0.018	0.868 ± 0.019	0.881 ± 0.023	0.871 ± 0.020	0.865 ± 0.019	0.864 ± 0.018	0.893 ± 0.030
3	0.798 ± 0.018	0.743 ± 0.033	0.699 ± 0.071	0.792 ± 0.017	0.801 ± 0.014	0.813 ± 0.016	0.825 ± 0.024
4	0.765 ± 0.088	0.731 ± 0.091	0.811 ± 0.109	0.739 ± 0.074	0.830 ± 0.051	0.821 ± 0.049	0.900 ± 0.051
5	0.321 ± 0.109	0.280 ± 0.125	0.337 ± 0.127	0.337 ± 0.111	0.394 ± 0.105	0.311 ± 0.065	0.454 ± 0.110
6	0.234 ± 0.081	0.239 ± 0.101	0.191 ± 0.068	0.217 ± 0.085	0.298 ± 0.063	0.322 ± 0.074	0.326 ± 0.074
7	0.667 ± 0.042	0.633 ± 0.069	0.686 ± 0.053	0.641 ± 0.061	0.725 ± 0.048	0.729 ± 0.054	0.747 ± 0.052
8	0.689 ± 0.022	0.688 ± 0.080	0.896 ± 0.074	0.697 ± 0.021	0.937 ± 0.018	0.932 ± 0.016	0.941 ± 0.047
9	0.899 ± 0.021	0.895 ± 0.053	0.874 ± 0.055	0.902 ± 0.024	0.887 ± 0.041	0.894 ± 0.037	0.907 ± 0.042
10	0.837 ± 0.044	0.792 ± 0.083	0.745 ± 0.113	0.846 ± 0.044	0.802 ± 0.053	0.812 ± 0.043	0.874 ± 0.058
11	0.812 ± 0.012	0.821 ± 0.060	0.812 ± 0.098	0.815 ± 0.016	0.824 ± 0.023	0.818 ± 0.000	0.939 ± 0.051
12	0.296 ± 0.076	0.336 ± 0.119	0.598 ± 0.096	0.311 ± 0.077	0.639 ± 0.077	0.652 ± 0.062	0.657 ± 0.077
13	0.492 ± 0.076	0.506 ± 0.084	0.319 ± 0.120	0.507 ± 0.073	0.439 ± 0.027	0.430 ± 0.025	0.528 ± 0.085
14	0.827 ± 0.020	0.824 ± 0.041	0.771 ± 0.053	0.829 ± 0.018	0.854 ± 0.016	0.854 ± 0.018	0.878 ± 0.031
15	0.631 ± 0.072	0.590 ± 0.162	0.578 ± 0.150	0.614 ± 0.060	0.496 ± 0.119	0.474 ± 0.092	0.685 ± 0.106
Mean	0.639	0.631	0.650	0.636	0.687	0.682	0.759
p-value	3.052E-05	3.052E-05	3.052E-05	3.052E-05	3.052E-05	3.052E-05	—

improvements are statistically significant. Collectively, these findings underscore the effectiveness of the proposed PAPSO algorithm. Its problem-aware design enables robust adaptation to different dataset characteristics, leading to highly competitive performance and highlighting its potential for practical applications in high-dimensional feature selection.

5.4. Running time

To provide a more comprehensive evaluation, this section compares the running time of PAPSO with other baseline algorithms. The detailed results are presented in Table 8. Although SFE-PSO consistently exhibits the lowest computational cost across all datasets,

Table 7

Performance comparison across different algorithms with F1-score.

No.	CUS-SPSO	SFE-PSO	MGWO	EBPSO	SSDR	FPSO	PAPSO
1	0.591 ± 0.043	0.599 ± 0.110	0.544 ± 0.095	0.584 ± 0.048	0.510 ± 0.033	0.487 ± 0.036	0.818 ± 0.061
2	0.894 ± 0.015	0.887 ± 0.015	0.878 ± 0.024	0.890 ± 0.017	0.862 ± 0.019	0.861 ± 0.019	0.890 ± 0.031
3	0.801 ± 0.019	0.750 ± 0.036	0.682 ± 0.088	0.794 ± 0.016	0.800 ± 0.014	0.812 ± 0.016	0.824 ± 0.024
4	0.835 ± 0.082	0.766 ± 0.096	0.805 ± 0.114	0.807 ± 0.037	0.827 ± 0.060	0.817 ± 0.049	0.898 ± 0.052
5	0.436 ± 0.153	0.373 ± 0.204	0.333 ± 0.121	0.443 ± 0.148	0.365 ± 0.111	0.259 ± 0.062	0.435 ± 0.109
6	0.252 ± 0.120	0.261 ± 0.125	0.175 ± 0.066	0.234 ± 0.117	0.282 ± 0.067	0.303 ± 0.074	0.311 ± 0.072
7	0.695 ± 0.041	0.661 ± 0.078	0.687 ± 0.054	0.663 ± 0.062	0.729 ± 0.046	0.732 ± 0.054	0.750 ± 0.050
8	0.705 ± 0.021	0.706 ± 0.070	0.895 ± 0.074	0.713 ± 0.015	0.936 ± 0.018	0.932 ± 0.016	0.940 ± 0.043
9	0.930 ± 0.011	0.918 ± 0.045	0.877 ± 0.053	0.931 ± 0.013	0.894 ± 0.036	0.901 ± 0.034	0.909 ± 0.043
10	0.881 ± 0.032	0.818 ± 0.098	0.725 ± 0.132	0.890 ± 0.032	0.761 ± 0.073	0.779 ± 0.052	0.876 ± 0.051
11	0.820 ± 0.015	0.832 ± 0.062	0.809 ± 0.099	0.825 ± 0.023	0.802 ± 0.027	0.793 ± 0.000	0.940 ± 0.051
12	0.594 ± 0.186	0.592 ± 0.223	0.577 ± 0.102	0.675 ± 0.196	0.612 ± 0.087	0.641 ± 0.062	0.638 ± 0.083
13	0.572 ± 0.110	0.568 ± 0.121	0.305 ± 0.116	0.584 ± 0.105	0.400 ± 0.018	0.396 ± 0.014	0.515 ± 0.092
14	0.841 ± 0.017	0.837 ± 0.034	0.772 ± 0.052	0.841 ± 0.015	0.855 ± 0.016	0.855 ± 0.013	0.878 ± 0.031
15	0.643 ± 0.090	0.614 ± 0.189	0.582 ± 0.126	0.619 ± 0.063	0.532 ± 0.109	0.509 ± 0.032	0.693 ± 0.104
Mean	0.699	0.679	0.643	0.699	0.678	0.672	0.754
p-value	1.077E-02	7.629E-04	3.052E-05	2.063E-02	3.052E-05	6.104E-04	—

Table 8

Running time of all algorithms.

No.	CUS-SPSO	SFE-PSO	MGWO	EBPSO	SSDR	FPSO	PAPSO
1	465.450	30.463	72.101	119.111	214.260	115.141	106.980
2	497.762	43.377	104.880	130.066	268.539	154.124	155.340
3	2218.769	384.077	767.456	1217.107	2284.125	1235.239	593.391
4	404.054	26.725	117.957	104.608	125.400	112.424	70.649
5	427.752	39.182	127.760	101.761	121.412	105.757	97.390
6	432.469	25.736	129.464	97.319	120.940	105.828	92.778
7	522.430	43.421	206.706	169.047	266.876	189.840	145.871
8	464.190	43.385	137.332	133.341	240.772	152.949	78.198
9	463.169	27.479	137.181	119.371	166.954	139.692	80.091
10	427.651	35.297	141.522	113.494	136.523	131.370	79.701
11	590.791	27.327	130.992	106.754	136.809	124.144	71.576
12	692.611	26.755	126.362	106.147	127.996	116.757	66.947
13	766.527	26.380	149.163	125.996	135.598	153.657	113.607
14	968.312	48.590	205.155	257.217	407.833	286.211	160.437
15	791.126	23.306	144.616	86.693	112.158	95.810	69.050

its overall performance is significantly inferior to our proposed method PAPSO. The primary reason for PAPSO's slightly higher computational cost compared to SFE-PSO is its incorporation of two feature evaluation steps: calculating ReliefF and MI. Importantly, PAPSO achieves competitive efficiency, requiring less running time than most comparison algorithms on the majority of datasets — specifically, it ranks as the second fastest method on 13 out of 15 datasets. This demonstrates that PAPSO offers a favorable trade-off between optimization quality and computational cost.

5.5. Ablation analysis

5.5.1. Effect of proposed mechanisms

To validate the effectiveness of the proposed mechanisms, this section details an ablation study comparing PAPSO with two specifically designed variants: PAPSOa and PAPSOB. PAPSOa replaces the proposed dataset-specific acceleration coefficient initialization strategy with traditional fixed coefficients. PAPSOB adopts a standard fixed inertia weight instead of the proposed inertia weight adjustment via PSO mechanism. The comparative performance results are presented in Table 9. Analyzing the primary classification accuracy, PAPSO achieves the best performance compared to both PAPSOa and PAPSOB on 9 out of 15 datasets. This finding strongly indicates that both the inertia weight adjustment and the dataset-specific acceleration coefficient initialization significantly contribute to PAPSO's overall effectiveness. While PAPSO occasionally selects more features than its variants, this reflects its inherent design focus: prioritizing classification accuracy through effective feature representation, rather than solely minimizing subset size. This demonstrates a strategic trade-off well-suited for complex, high-dimensional scenarios. Collectively, this ablation study validates the individual and combined contributions of PAPSO's problem-aware hyperparameter design components.

5.5.2. Effect of combination of ReliefF and MI

This section investigates the effectiveness of combining ReliefF and MI within PAPSO via an ablation study. The full PAPSO algorithm is compared against three specific settings: one variant using only the ReliefF metric (Only ReliefF), another variant using only the MI metric (Only MI), and a standard Baseline PSO which operates using the velocity update mechanism described in this

Table 9
Comparison of accuracy and feature size across different variants.

No.	PAPSOa		PAPSOb		PAPSO	
	Acc	Size	Acc	Size	Acc	Size
1	0.826	43.067	0.774	28.933	0.825	55.000
2	0.887	78.233	0.906	42.000	0.893	72.400
3	0.813	53.733	0.820	67.500	0.825	97.133
4	0.864	11.933	0.855	9.633	0.900	9.867
5	0.415	122.467	0.422	298.067	0.454	298.100
6	0.280	100.667	0.331	108.300	0.326	130.633
7	0.713	157.267	0.737	291.367	0.747	301.100
8	0.951	8.400	0.942	9.500	0.941	8.267
9	0.897	7.400	0.884	9.667	0.907	8.333
10	0.836	7.033	0.855	7.233	0.874	3.600
11	0.956	5.833	0.920	5.333	0.939	4.833
12	0.604	93.867	0.635	81.333	0.657	82.600
13	0.533	103.367	0.509	78.100	0.528	61.567
14	0.854	150.900	0.852	179.633	0.878	175.000
15	0.663	14.700	0.670	11.433	0.685	11.200
Mean	0.739	63.924	0.741	81.869	0.759	87.976

Table 10
Comparison of accuracy and feature size across different metric combinations.

No.	Baseline PSO		Only ReliefF		Only MI		PAPSO	
	Acc	Size	Acc	Size	Acc	Size	Acc	Size
1	0.493	1578.200	0.748	12.241	0.799	27.379	0.825	55.000
2	0.867	1528.433	0.870	26.759	0.889	49.069	0.893	72.400
3	0.801	2511.433	0.817	26.655	0.801	208.897	0.825	97.133
4	0.806	3463.600	0.892	6.724	0.856	9.862	0.900	9.867
5	0.339	3946.567	0.375	43.103	0.370	593.621	0.454	298.100
6	0.265	3943.567	0.268	25.759	0.284	135.103	0.326	130.633
7	0.716	4218.300	0.700	48.414	0.717	255.275	0.747	301.100
8	0.935	3738.033	0.943	5.207	0.929	5.483	0.941	8.267
9	0.897	4597.867	0.870	5.793	0.889	4.103	0.907	8.333
10	0.788	4531.167	0.840	4.345	0.853	2.862	0.874	3.600
11	0.818	4542.567	0.903	5.034	0.895	4.759	0.939	4.833
12	0.733	4832.867	0.598	6.207	0.603	63.897	0.657	82.600
13	0.422	6176.867	0.423	28.828	0.469	74.862	0.528	61.567
14	0.854	6697.800	0.829	53.621	0.838	149.517	0.878	175.000
15	0.489	7902.633	0.602	3.333	0.655	5.214	0.685	11.200
Mean	0.682	4280.687	0.712	20.1349	0.723	106.060	0.759	87.976

paper, without ReliefF/MI guidance. Detailed comparative results are presented in Table 10. As shown, the full PAPSO algorithm achieves the best performance, based on classification accuracy, on 14 out of 15 datasets compared to these alternatives. Furthermore, PAPSO demonstrates a significantly higher average classification accuracy across all datasets than Baseline PSO, Only ReliefF, and Only MI. Regarding feature selection parsimony, PAPSO also generally selects fewer features on average compared to two comparison settings. These results strongly validate the effectiveness of incorporating both ReliefF and MI in PAPSO. The findings indicate that integrating these complementary metrics allows PAPSO to strike a superior balance between local relevance and global relevance, ultimately leading to improved feature selection performance compared to using none or only one metric.

5.6. Parameter sensitivity analysis

In Section 3.2, the acceleration coefficients, after initialization, are mapped to a predefined interval $[c_{min}, c_{max}]$. To assess the sensitivity of the proposed PAPSO algorithm to the specific choice of this interval, we conduct a parameter sensitivity analysis. Drawing inspiration from related works [16,23] which suggest typical bounds, and aiming for a comprehensive evaluation, we tested multiple $[c_{min}, c_{max}]$ combinations derived from $c_{min} \in [0.5, 1]$ and an extended range for $c_{max} \in [1.5, 3]$. Specifically, the combinations evaluated are $[0.5, 1.5]$, $[0.5, 2]$, $[0.5, 2.5]$, $[0.5, 3]$, $[1, 1.5]$, $[1, 2]$, $[1, 2.5]$, $[1, 3]$. The detailed performance impact of varying $[c_{min}, c_{max}]$ settings is further explored across individual datasets, as visualized in Fig. 4. In this figure, each box plot corresponds to one specific dataset, illustrating the distribution of fitness values achieved by PAPSO under the eight different $[c_{min}, c_{max}]$ configurations tested. The number shown above each box indicates the standard deviation of these fitness values. The relatively compact nature of the individual box plots in Fig. 4, combined with the uniformly small standard deviations annotated above them, clearly indicates that the algorithm's performance exhibits limited variation across the different $[c_{min}, c_{max}]$ settings for each dataset. Complementing this visual analysis, Table 11 provides quantitative evidence by summarizing the mean fitness values averaged across all datasets for

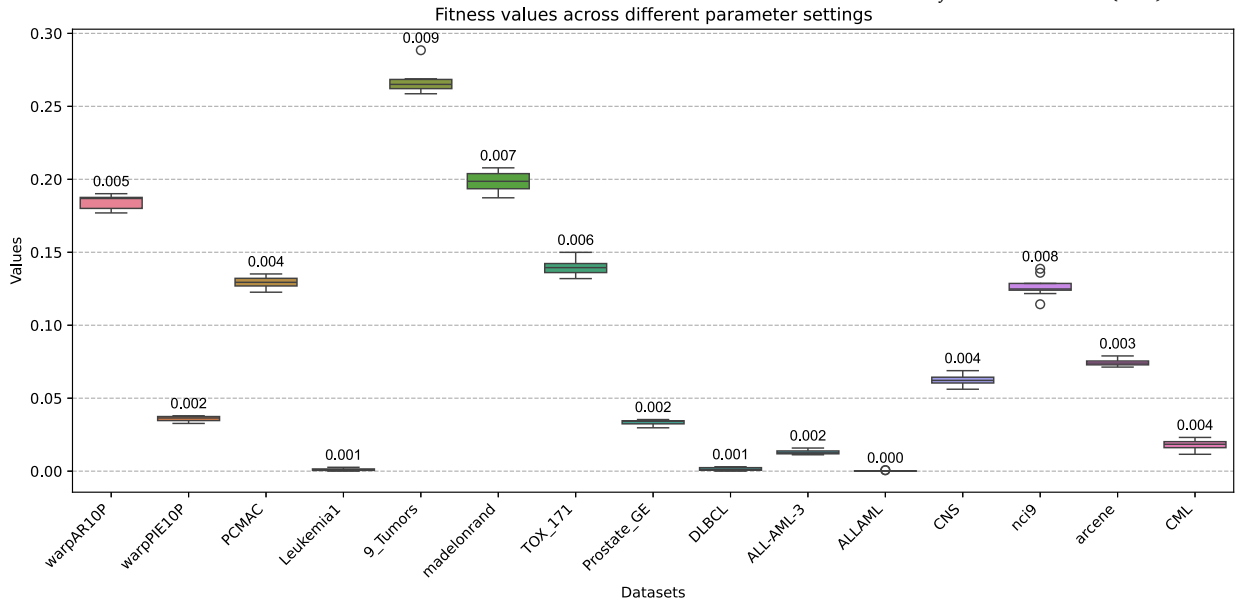


Fig. 4. Box-whisker-plots with different settings.

Table 11

The mean fitness values across all datasets with different settings.

	[0.5, 1.5]	[0.5, 2]	[0.5, 2.5]	[0.5, 3]	[1, 1.5]	[1, 2]	[1, 2.5]	[1, 3]
Mean	0.084	0.085	0.086	0.084	0.085	0.087	0.088	0.087

Table 12

The classification accuracy with different feature subsets.

	F_{1151}	F_{2298}	F_{8773}	F_{12555}	None
Acc	0.778	0.778	0.667	0.778	0.889

each parameter configuration. As previously noted and clearly shown in Table 11, these average performance metrics are remarkably stable, varying only slightly between 0.084 and 0.088 across all eight distinct settings. Considering both the visual evidence of limited performance fluctuation within individual datasets (Fig. 4) and the quantitative stability of the average performance across all datasets and settings (Table 11), it strongly indicates that the proposed PAPSO algorithm is largely insensitive to the specific selection of c_{min} and c_{max} within the evaluated bounds.

5.7. Further analysis

FS aims to identify relevant features while eliminating irrelevant and redundant ones. In this section, we validate the effectiveness of the PAPSO algorithm in achieving this goal using the high-dimensional CML dataset. Based on 30 independent runs of PAPSO on the CML dataset, we generated a top feature frequency chart that summarizes the unique feature subsets discovered. Fig. 5 illustrates that the top 5 most frequently selected features are F_{12555} , F_{6871} , F_{9149} , F_{9392} , F_{9161} . Furthermore, each row represents a feature subset, with the leftmost column showing the number of features in that subset, and the rightmost column showing the corresponding classification accuracy. Notably, F_{12555} is selected in 11 out of 15 individuals, and the subsets containing these high-frequency features consistently achieved very high accuracy, which preliminarily indicates the significant importance of the features chosen by PAPSO. Due to the lack of specific feature descriptions in the CML dataset, we further evaluated the importance ranking of these five frequently selected features using the ReliefF and MI criteria. For instance, under the ReliefF metric, F_{12555} ranks the highest, while under the MI metric, F_{6871} achieves the second rank. These findings not only emphasize the importance of the features selected by the PAPSO algorithm, but also confirm its ability to effectively integrate and leverage both ReliefF and MI evaluation criteria.

Furthermore, to investigate potential redundancy within the selected feature sets, we select a feature subset obtained from the experiments on the CML dataset, which achieved the highest classification accuracy with a relatively small number of features. This subset comprises features F_{1151} , F_{2298} , F_{8773} and F_{12555} . We employ a leave-one-out validation strategy, sequentially removing each feature from this subset and evaluating the classification accuracy using the remaining features. The detailed results are presented in



Fig. 5. Top feature frequency chart (According to ReliefF, F_{12555} is the most important feature. According to MI, F_{6871} is the second important feature).

Table 12. The results clearly demonstrate that the complete five-feature subset yields the highest classification accuracy. Crucially, the removal of any single feature leads to a decrease in accuracy. This outcome strongly indicates that the features within this minimal subset are complementary and non-redundant, further highlighting PAPSO's effectiveness in identifying concise sets of essential features while discarding irrelevant and redundant ones.

6. Conclusion

In this paper, we introduce PAPSO, a novel Particle Swarm Optimization algorithm tailored for high-dimensional feature selection. PAPSO embodies a problem-aware hyperparameter design strategy by effectively leveraging dataset-specific insights from ReliefF and MI metrics. Specifically, PAPSO features two key innovations: (1) an inertia weight adjustment mechanism where the crucial balance (α_1, α_2) between ReliefF and MI contributions is dynamically tuned via the PSO process itself, and (2) a dataset-specific initialization strategy for acceleration coefficients, assigning unique initial $c_1(j)$ and $c_2(j)$ values to each feature based on its ReliefF and MI statistics respectively. To enhance search robustness, PAPSO also incorporates a random perturbation element into its search operator. Experiments on 15 high-dimensional datasets (up to 12,625 dimensions) demonstrate PAPSO's effectiveness. It significantly outperforms peers and variants, achieving a strong balance between feature count and classification accuracy (often prioritizing accuracy), thereby validating its problem-aware components, particularly in high dimensions. Further detailed analysis confirms PAPSO's ability to identify relevant features and remove redundant ones.

For future work, enhancing scalability for extremely high-dimensional problems is a key direction. This could involve incorporating strategies into PAPSO that reduce the effective dimensionality of the search space explored by the optimizer, such as feature pre-screening or search space decomposition techniques. However, a key scalability challenge for PAPSO is that the computational overhead of ReliefF and MI grows significantly with the number of samples. Therefore, an important avenue for future research is to optimize their implementation—for instance, through parallel computing—to substantially reduce this cost. Improving PAPSO's efficiency in this way would make it practical to integrate into deep learning pipelines as a data preprocessing step, ultimately enhancing model performance.

CRedit authorship contribution statement

Jinrui Gao: Writing – original draft, Software, Methodology, Conceptualization. **Zhenyu Lei:** Writing – review & editing, Software, Methodology, Conceptualization. **Tao Zheng:** Supervision, Conceptualization. **Lijun Guo:** Writing – review & editing, Supervision, Conceptualization. **Yirui Wang:** Writing – review & editing, Supervision, Conceptualization. **Shangce Gao:** Writing – review & editing, Supervision, Software, Methodology, Conceptualization.

Declaration of competing interest

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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Data availability

Data will be made available on request.

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